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1 資料結構

1.1 BIT

```

1 struct Binary_Indexed_Tree{
2     int n;
3     vector<int> bit;
4     void init(int _n){
5         n=_n+1;
6         bit=vector<int>(n,0);
7     }
8     int lowbit(int x) return x&-x;
9     void update(int x,int v){
10         for(;x<n;x+=lowbit(x)){
11             bit[x]+=v;
12         }
13     void range_update(int l,int r,int v){
14         update(l,v);
15         update(r+1,-v);
16     }
17     int qry(int x){
18         int ans=0;
19         for(;x>0;x-=lowbit(x)){
20             ans+=bit[x];
21         }
22         return ans;
23 };

```

2 字串

2.1 字典樹 trie

```

1 struct trie{
2     trie *nxt[26]={};
3     int cnt=0;
4     int sz=0;
5 };
6 trie *root=new trie();
7 void insert(const string &s){
8     trie *now=root;
9     for(auto i:s){
10         if(now->nxt[i-'a']==NULL){
11             now->nxt[i-'a']=new trie();
12         }
13         now=now->nxt[i-'a'];
14     }
15 int qry(string &s){
16     trie *now=root;
17     for(auto i:s){
18         if(now->nxt[i-'a']==NULL) return;
19         now=now->nxt[i-'a'];
20     }
21     return now->cnt;
22 }

```

2.2 KMP

```

1 vector<int> kmp(string s,string t){
2     if(t.size()>s.size()) return {};
3     vector<int> pi(t.size(),0);
4     for(int i=1,j=0;i<t.size();i++){
5         while(j>0 && t[j]!=t[i]){
6             j=pi[j-1];
7         }
8         if(t[j]==t[i]) j++;
9         pi[i]=j;
10    }
11    vector<int> ans;
12    for(int i=0,j=0;i<s.size();i++){
13        while(j>0 && t[j]!=s[i]){
14            j=pi[j-1];
15        }
16        if(t[j]==s[i]) j++;
17
18        if(j==t.size()){
19            ans.push_back(i-j+1);
20            j=pi[j-1];
21        }
22    }
23    return ans;
}

```

3 最短路徑

3.1 basic

```

1 // c++ code
2 #include <bits/stdc++.h>
3 using namespace std;
4
5 int main() {
6     // test comment
7     cout << "test string\n";
8 }

```

4 圖論

4.1 凸包問題

```

1 struct node{
2     int x,y;
3     bool operator<(const node &other) const{
4         return
5             (x<other.x) || (x==other.x && y<other.y);
6     };
7 };
8 struct ConvexHull{
9     vector<node> vec;
10    int n;
11    void init(vector<node> &v){
12        vec=v;
13        n=v.size();
14    }
15    int cross(node a,node o,node b){
16        return
17            (a.x-o.x)*(b.y-o.y)-(b.x-o.x)*(a.y-o.y);
18    }
19    vector<int> point;
20    vector<int> area;
21    void build(){
22        point=vector<int>(n+1);
23        area=vector<int>(n+1);
24        vector<node> up,down;

```

```

25     int upsum=0;
26     int downsum=0;
27     for(int i=0;i<n;i++){
28         while(up.size()>=2 &&
29             cross(up[up.size()-2],
30                 up[up.size()-1],
31                 vec[i])<=0){
32             upsum-=cross(up[up.size()-2],
33                         {0,0},up[up.size()-1]);
34             up.pop_back();
35         }
36         up.push_back(vec[i]);
37         if(up.size()>1)
38             upsum+=cross(up[up.size()-2],
39                         {0,0},
40                         up[up.size()-1]);
41
42         while(down.size()>=2 &&
43             cross(down[down.size()-2],
44                 down[down.size()-1],
45                 vec[i])>=0){
46             downsum-=cross(down[down.size()-2],
47                           {0,0},down[down.size()-1]);
48             down.pop_back();
49         }
50         down.push_back(vec[i]);
51         if(down.size()>1)
52             downsum+=cross(down[down.size()-2],
53                           {0,0},down[down.size()-1]);
54
55         area[i+1]=abs(upsum-downsum);
56
57         if(i+1==1) point[i+1]=1;
58         else point[i+1]=up.size()+down.size()-2;
59     }}
60
61     double getarea(int p){
62         return area[p]/2.0;
63     }
64     int getnum(int p){
65         return point[p];
66     }
67 };

```

- Kirchhoff's theorem :
 $A_{ii} = \deg(i), A_{ij} = (i, j) \in E ? -1 : 0$, Deleting any one row, one column, and cal the $\det(A)$
- Polya' theorem (c is number of color , m is the number of cycle size):
 $(\sum_{i=1}^m c^{gcd(i,m)})/m$
- Burnside lemma:
 $|X/G| = \frac{1}{|G|} \sum_{g \in G} |X^g|$
- 錯排公式: (n 個人中, 每個人皆不再原來位置的組合數):
 $dp[0] = 1; dp[1] = 0;$
 $dp[i] = (i-1) * (dp[i-1] + dp[i-2]);$
- Bell 數 (有 n 個人, 把他們拆組的方法總數) :
 $B_0 = 1$
 $B_n = \sum_{k=0}^n s(n, k) \quad (\text{second-stirling})$
 $B_{n+1} = \sum_{k=0}^n \binom{n}{k} B_k$
- Wilson's theorem :
 $(p-1)! \equiv -1 \pmod{p}$
- Fermat's little theorem :
 $a^p \equiv a \pmod{p}$
- Euler's totient function:
 $A^{B^C} \pmod{p} = \text{pow}(A, \text{pow}(B, C, p-1)) \pmod{p}$
- 歐拉函數降幂公式:
 $A^B \pmod{C} = A^{B \pmod{\phi(C) + \phi(C)}} \pmod{C}$
- 6 的倍數:
 $(a-1)^3 + (a+1)^3 + (-a)^3 + (-a)^3 = 6a$
- Standard young tableau (標準楊表):
 $\lambda = (\lambda_1 \geq \dots \geq \lambda_k), \sum \lambda_i = n$ denoted by $\lambda \vdash n$
 $\lambda \vdash n$ 意思為 λ 整數拆分 n eg. $n = 10, \lambda = (6, 4)$ 此拆分可表示一種楊表形狀。
 楊表: 第 1 列 λ_1 行 $\dots$ 第 k 列 λ_k 行的方格圖。
 標準楊表: 每列從左到右遞增, 每行從上到下遞增。
 Let T 為某一 Permutation 跑 RSK 後的標準楊表, 則此 Permutation 的 LDS、LIS 長度分別為 T 的列、行數。
- RSK Correspondence:
 A permutation is bijective to (P, Q) 一對標準楊表
 P : Permutation 跑 RSK 算法的結果, 可為半標準楊表。
 Q : 可用來還原 Permutation (像排列矩陣)。
- Hook length formula (形狀為 λ 的標準楊表個數):
 $f^\lambda = \frac{n!}{\prod h_{\lambda(i,j)}}$
 $h_{\lambda(i,j)}$ = number of pair (x, y) where $(x = i \vee y = j) \wedge (x, y) \geq (i, j)$ 且 (x, y) 落在形狀為 λ 的表上。
 Recursion:
 (i) $f^{(0, \dots, 0)} = 1$
 (ii) $f^{(\lambda_1, \dots, \lambda_m)} = \sum_{k=1}^m f^{(\lambda_1, \dots, \lambda_{k-1}, \lambda_k-1, \lambda_{k+1}, \dots, \lambda_m)}$
- 一個環上相鄰顏色不同的染色數量:
 $f(n, k) = (k-1)^n + (-1)^n (k-1)$
 n 是點的數量, k 是顏色的數量

5 數學

5.1 公式

- Lucas's Theorem :
 For $n, m \in \mathbb{Z}^*$ and prime P , $C(m, n) \pmod{P} = \prod C(m_i, n_i)$ where m_i is the i -th digit of m in base P .
- Stirling approximation :
 $n! \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n e^{\frac{1}{12n}}$
- Stirling Numbers(permutation $|P| = n$ with k cycles):
 $S(n, k) = \text{coefficient of } x^k \text{ in } \Pi_{i=0}^{n-1} (x+i)$
- Stirling Numbers(Partition n elements into k non-empty set):
 $S(n, k) = \frac{1}{k!} \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n$
- Pick's Theorem : $A = i + b/2 - 1$
 A : Area
 i : grid number in the inner
 b : grid number on the side
- Catalan number : $C_n = \binom{2n}{n} / (n+1)$
 $C_n^{n+m} - C_{n+1}^{n+m} = (m+n)! \frac{n-m+1}{n+1}$ for $n \geq m$
 $C_n = \frac{1}{n+1} \binom{2n}{n} = \frac{(2n)!}{(n+1)!n!}$
 $C_0 = 1$ and $C_{n+1} = 2 \binom{2n+1}{n+2} C_n$
 $C_0 = 1$ and $C_{n+1} = \sum_{i=0}^n C_i C_{n-i}$ for $n \geq 0$
- Euler Characteristic:
 planar graph: $V - E + F - C = 1$
 convex polyhedron: $V - E + F = 2$
 V, E, F, C : number of vertices, edges, faces(regions), and components

[illegible]